Stats - Sampling (2)

Systematic Sampling

- **Definition and Context**: Systematic sampling involves selecting a sample from a population where every kth individual is chosen after a random start.
- **Population Size**: The population size is denoted by a large (N).
- **Sample Size**: A sample size is determined; in this case, it is (n = 120).
- **Calculation of k**:
    - The value (k) is calculated as (k = \frac{N}{n} = \frac{200}{120} = 1.67) (rounded to a whole number, should be treated as intervals of 10 based on the context).
- **Random Number Generation**: The first step after determining (k) is to select a random number between 1 and (k).
    - Example: If the random number generated is 7, this indicates the starting point for selection.
- **Selection Process**:
    - In the first block (1 to (k = 10)), select 7.
    - In subsequent blocks, continue to add (k):
    - 2nd block: 7 + 10 = 17
    - 3rd block: 17 + 10 = 27
    - 4th block: 27 + 10 = 37
    - Continue this process until 120 items are selected.
- **Summary of Mechanism**:
    - The importance of the value of (k) is twofold:
    1. It helps in generating the random starting number.
    2. It allows systematic selection by adding (k) to the starting point for subsequent selections.

Stratified Sampling

- **Definition**: Stratified sampling involves dividing the population into strata or groups that share similar characteristics and sampling accordingly.
- **Key Characteristics**: It is crucial when certain population characteristics must be reflected in the sample to maintain representativeness.
- **Example of Strata**:
    - Consider the characteristic of resident status: either students in residence (res) or those with private accommodations.
- **Population Composition**: For the selected group, assume a 50/50 split.

    - Example: 600 in res, and 600 in private accommodation.
    - Total population = 1200.
- **Selection Protocol**:
    - Split the population into two strata (residents and private).
    - From each stratum, sample accordingly (e.g., 10% from each).
    - For residents: 10% of 600 = 60 students.
    - For private accommodation: 10% of 600 = 60 students.
- **Resulting Sample Composition**:
    - Sample of 120 yields 60 from res and 60 from private accommodation, thus maintaining the original population split.
- **Purpose**: This method ensures that important characteristics are preserved within the sample, which could be lost through random sampling alone.
- **Practical Consideration**:
    - In the digital age, organizational lists can help to conduct stratified sampling efficiently without necessitating a physical split.

Cluster Sampling

- **Definition**: Cluster sampling uses naturally occurring groups (clusters) from a population, and the whole groups are sampled rather than individuals.
- **Key Feature**: The groups should represent a fair variation of the population's characteristics.
- **Example**:
    - Consider a stats module with 2000 students divided into groups of roughly 200 each.
- **Sampling Protocol**:
    - Instead of sampling all 2000 students, select just two groups or clusters.
    - Thus, if clusters 1 and 2 are chosen, the sample size becomes 400.
- **Advantages**:
    - Faster and cheaper compared to alternative methods, as it leverages pre-existing groups.
- **Comparison with Other Sampling Methods**:
    - Unlike stratified sampling, which ensures representation from each defined strata, cluster sampling accepts existing groups allowing for a broader sampling at a reduced cost.

Weighted Sampling of Invoices

- **Context**: The process involves sampling invoices based on their total monetary value instead of equal likelihood.
- **Notation**:
    - Let (T) denote the total monetary value of all invoices.
    - For this example, assume (T = 225,000).

- **Population Size**:
  - Let (m = 1500), the total number of invoices in the population.
- **Sample Size**:
  - The target sample size is (n = 20).
- **Random Number Generation Process**:
  - Generate random numbers between 1 and (T) (here, between 1 and 225000) to select invoices for sampling.
- **Setting Up a Table**:
  - Include columns such as invoice number and value; ensure that the total values align with sampled amounts.
- **Accumulated Value Calculation**:
  - Create an accumulating value table that adds invoice values across rows.
  - The first entry equals 10 (assuming the first value is 10), then add each subsequent invoice value to accumulate until all entries are filled.
- **Creating Intervals**:
  - Define intervals from the accumulated values taking care to avoid overlaps. Example:
  - First interval: ([1, 10])
  - Second interval: ([11, 122]) (skipping overlap of 10)
  - Continue creating further intervals based on the accumulated total values.
- **Random Number Assignment**:
  - Assign generated random numbers to respective intervals (e.g., 98 falls in the second interval).
- **Sample Selection Outcome**:
  - If a random number falls into an interval corresponding to an already chosen invoice, disregard and continue to new selections until 20 unique invoices are attained.
- **Probability of Selection**:
  - To define the probability of selecting an envelope with value (x):
  - The probability = (P(x) = \frac{value(x)}{T})
- **Purpose of the Process**:
  - Aim is to focus sampling on higher monetary value invoices to prioritize auditing significant entries.

Data Quality and Cleanup (''Dirty Data'')

- **Definition**: Dirty data consists of inaccuracies or errors within a data set that affect integrity and quality.
- **Identification of Issues**:
  - Analyze a dataset and identify discrepancies, such as missing values in rows that contribute to aggregate values.
- **Example of Issues**:

    - Total computed counts and manually added values may not align; analyze through counts.
- **Common Causes of Dirty Data**:
    - Human error, incorrect data entry methods, lack of validation procedures during collection processes.